Natanael S. Alpay

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SUMMARY

Mathematics Ph.D. candidate with 10+ publications and preprints in analysis, probability, and machine learning. Uses LLMs as conjecture engines, then verifies candidate results through mathematical proof and reproducible computation. Develops and evaluates neural hedging strategies in Python/PyTorch and studies portfolio compression, with quantitative finance training through the Erdős Institute. Applied work also includes healthcare signal prioritization. Has presented over 20 research talks. Seeking quantitative research roles in finance.

EDUCATION

Ph.D. Mathematics	University of California, Irvine	(GPA: 3.720)	**Expected 2027**
M.S. Applied Mathematics	University of California, Irvine	(GPA: 3.720)	**2025**
B.S. Mathematics	Chapman University	(GPA: 3.849)	**2022**
B.S. Physics	Chapman University	(GPA: 3.849)	**2022**
B.S. Computer Science	Chapman University	(GPA: 3.849)	**2022**
B.A. French	Chapman University	(GPA: 3.849)	**2022**

- Earned the Ronald M. Huntington Outstanding Scholarship Award, Outstanding mathematics and physics student award

SKILLS

- Programming: Python (PyTorch, NumPy, pandas, Matplotlib), SQL, R, C/C++, JavaScript, MATLAB,Mathematica.
- Machine Learning: Neural networks, SGD, logistic regression, gradient boosting, temporal validation, calibration.
- Quantitative Methods: Stochastic processes, optimization, Monte Carlo simulation, numerical analysis, kernel methods.
- Languages: English, French, Russian, Hebrew.
- Finance training: Erdős Institute Quant Finance Boot Camp: option pricing, Monte-Carlo, hedging, and backtesting.

SELECTED PROJECTS

Memory-Efficient Deep Hedging [GitHub/Deep-Hedging-Implementation] **2026 - Present**

- Implemented Python/PyTorch neural hedging strategies in simulated Heston markets with transaction costs, trained under CVaR and entropic risk. On going project.
- Reduced peak memory usage by 63.8% through dynamic-batch training versus fixed-pool training in large portfolio experiment, with similar test CVaR and 29.4% lower entropic risk in a preliminary small portfolio experiment.

Complex Optimization and Kernel Regression for Inverse Problems (arXiv:2604.22165 | arXiv:2604.23017) **2026**

- Developed Python,C++ implementations of complex-valued gradient methods to reconstruct special solutions and study which solutions the algorithms learn, including inverse problems with initial conditions arising from quantum mechanics.
- Established convergence guarantees under stated assumptions and checked them through numerical experiments.

Machine Learning for Drug and Vaccine Signal Prioratization [GitHub/VAERS-Signal-Prioritization] **2025 - 2026**

- Built Python logistic-regression and gradient-boosting models to prioritize VAERS vaccine-symptom pairs. On a temporal holdout, captured 50% of eligible next-quarter threshold crossings with 47.4% fewer candidates for review. Presented the project at CHOC's Pediatric and Lifespan Data Science Conference (April 29, 2026) and Berkeley AIPM (October 2026).
- Refined Python gradient-boosting models for FAERS drug-safety signals by removing unstable temporal features, improving AUPRC by about 25%.

RESEARCH AND TEACHING EXPERIENCE

Graduate Research and Teaching Assistant - University of California, Irvine **Sep 2022 - Present**

- Investigate simpler portfolio summaries for managing risk through Deep Bellman Hedging, a current research direction.
- Working on sharp Gaussian-mixture stability bounds: how model misspecification affects score accuracy and sampling.
- Proved sharp bounds based on the stopping time of a stochastic process starting node, with calculations implemented in Python; tested ideas suggested by large language models through proofs and computational checks.
- Taught Numerical Analysis I and II as instructor of record (Summer 2026), supported mathematics courses as a teaching assistant, and mentored DATA@UCI students in Python analysis and presentations (Jan–Jun 2025).
- Lead the Deep Hedging Directed Reading Program, guiding undergraduates through Deep Hedging and Deep Bellman Hedging papers, Python experiments, and critical evaluation; presented this work to UCI's finance club.
- Mentored eight undergraduates on two random-matrix projects, guiding Python experiments and two research posters.

Machine Learning Research and Data Analysis- Chapman University **Jun 2020 - Sep 2022**

- Analyzed Brain Institute data using machine-learning models in Python and analysis tools in SQL and JavaScript.

SELECTED PUBLICATIONS AND PREPRINTS

Alpay & Battaglia. Complex Stochastic Gradient Descent and Directional Bias in Reproducing Kernel Hilbert Spaces. arXiv, 2026.
Alpay & Ivanisvili. Sharp Lower Bounds for Dyadic Square Functions of Indicator Functions of Sets. arXiv, 2025; revised 2026.
Alpay, Cerejeiras & Kähler. Generalized Fock-space and fractional derivatives: Uniqueness of sampling and interpolation sets. Results in Applied Mathematics, 2026.

Interests: Weightlifting, LEGO, karaoke.